Online Appendix for Asynchronous Split Federated Learning for Edge Systems with Partial Aggregation

This appendix provides the proofs omitted from the main manuscript.

A Proof to Lemma

For any round $k \in \{t-\tau_n(t)+1, \ldots, t\}$, the local server-side submodel of client n starts from the common server-side model, i.e.,

$$\boldsymbol{w}_{s,n}^{k,0} = \boldsymbol{w}_s^k. \tag{1}$$

After H_n^k server-side co-training steps, we have

$$\boldsymbol{w}_{s,n}^{k,H_n^k} - \boldsymbol{w}_s^k = -\eta \sum_{h=1}^{H_n^k} \boldsymbol{g}_{s,n}^{k,h}. \tag{2}$$

Therefore,

$$\begin{aligned}
\mathbb{E}\left[\left\|\boldsymbol{w}_{s,n}^{k,H_n^k} - \boldsymbol{w}_s^k\right\|^2\right] &= \eta^2\, \mathbb{E}\left[\left\|\sum_{h=1}^{H_n^k} \boldsymbol{g}_{s,n}^{k,h}\right\|^2\right] \\
&\leq \eta^2 H_n^k \sum_{h=1}^{H_n^k} \mathbb{E}\left[\left\|\boldsymbol{g}_{s,n}^{k,h}\right\|^2\right] \\
&\leq \eta^2 H_n^k \sum_{h=1}^{H_n^k} G_s^2 \\
&= \eta^2 (H_n^k)^2 G_s^2,
\end{aligned} \tag{3}$$

where the first inequality follows from

$$\left\|\sum_{h=1}^{H_n^k} \boldsymbol{g}_{s,n}^{k,h}\right\|^2 \leq H_n^k \sum_{h=1}^{H_n^k} \left\|\boldsymbol{g}_{s,n}^{k,h}\right\|^2. \tag{4}$$

Summing the above inequality over $k = t - \tau_n(t) + 1, \ldots, t$ yields

$$\Delta_{s,n}^t = \sum_{k=t-\tau_n(t)+1}^{t} \mathbb{E}\left[\left\|\boldsymbol{w}_{s,n}^{k,H_n^k} - \boldsymbol{w}_s^k\right\|^2\right] \leq \eta^2 G_s^2 \sum_{k=t-\tau_n(t)+1}^{t} (H_n^k)^2. \tag{5}$$

This completes the proof.

B Proof to Theorem

Since aggregation can only be performed with aligned submodels, we consider the aligned common submodels. We recall the global model at round t as:

$$\boldsymbol{x}^t = [\boldsymbol{w}_c^t; \boldsymbol{w}_s^t]. \tag{6}$$

The local model for client n at h-th step in round t is denoted as:

$$\boldsymbol{x}_n^{t,h} = [\boldsymbol{w}_{c,n}^{t-\tau_n(t)+1,h}; \boldsymbol{w}_{s,n}^{t,h}]. \tag{7}$$

We recall the dual-pace update of the server-side model $\boldsymbol{w}_s^t$ and the client-side model $\boldsymbol{w}_c^t$:

$$\begin{aligned} \boldsymbol{w}_s^{t+1} &= \sum_{n=1}^{N} a_n \boldsymbol{w}_{s,n}^{t,H_n^t} = \boldsymbol{w}_s^t - \eta^t \sum_{n=1}^{N} a_n \sum_{h=1}^{H_n^t} g_{s,n}^{t,h}, \\ \boldsymbol{w}_c^{t+1} &= \boldsymbol{w}_c^t - \eta^t \sum_{n\in\mathcal{M}_t} b_n^t \sum_{k=t-\tau_n(t)+1}^{t} \sum_{h=1}^{H_n^k} g_{c,n}^{k,h}. \end{aligned} \tag{8}$$

We first have the following lemma:

Lemma 1 (Convergence Decomposition)**.** *Under Assumption 1, we have:*

$$\begin{aligned} & F(\boldsymbol{x}^{t+1}) - F(\boldsymbol{x}^t) \\ \leq & \langle \nabla_{\boldsymbol{w}_s} F(\boldsymbol{x}^t), \boldsymbol{w}_s^{t+1} - \boldsymbol{w}_s^t \rangle + \frac{S}{2} \|\boldsymbol{w}_s^{t+1} - \boldsymbol{w}_s^t\|^2 \qquad (9) \\ + & \langle \nabla_{\boldsymbol{w}_c} F(\boldsymbol{x}^t), \boldsymbol{w}_c^{t+1} - \boldsymbol{w}_c^t \rangle + \frac{S}{2} \|\boldsymbol{w}_c^{t+1} - \boldsymbol{w}_c^t\|^2. \qquad (10) \end{aligned}$$

We refer to Proposition 3.5 in [1] to complete this lemma. We then bound the resulting two terms (9) and (10). We start from the server-side error bound in (9).

According to the aggregation scheme in (8), we obtain:

$$\mathbb{E}\left[\langle \nabla_{\boldsymbol{w}_s} F(\boldsymbol{x}^t), \boldsymbol{w}_s^{t+1} - \boldsymbol{w}_s^t \rangle\right] = \left\langle \nabla_{\boldsymbol{w}_s} F(\boldsymbol{x}^t), \mathbb{E}\left[-\eta^t \sum_{n=1}^{N} a_n \sum_{h=1}^{H_n^t} g_{s,n}^{t,h}\right] \right\rangle.$$

We notice that $1 = \sum_{n=1}^{N} a_n = \frac{\sum_{n=1}^{N} a_n H_n^t}{\sum_{n=1}^{N} a_n H_n^t}$. and $\langle a, b\rangle = \frac{1}{2}\|a\|^2 + \frac{1}{2}\|b\|^2 - \frac{1}{2}\|a-b\|^2$. Therefore, we obtain:

$$\begin{aligned} & \mathbb{E}\left[\langle \nabla_{\boldsymbol{w}_s} F(\boldsymbol{x}^t), \boldsymbol{w}_s^{t+1} - \boldsymbol{w}_s^t \rangle\right] \\ = & -\frac{\eta^t \sum_{n=1}^{N} a_n H_n^t}{2} \|\nabla_{\boldsymbol{w}_s} F(\boldsymbol{x}^t)\|^2 - \frac{\eta^t}{2\sum_{n=1}^{N} a_n H_n^t} \mathbb{E}\left[\left\| \sum_{n=1}^{N} a_n \sum_{h=1}^{H_n^t} \nabla_{\boldsymbol{w}_s} F_n(\boldsymbol{x}_n^{t,h}) \right\|^2\right] \\ & + \frac{\eta^t}{2\sum_{n=1}^{N} a_n H_n^t} \mathbb{E}\left[\left\| \sum_{n=1}^{N} a_n \sum_{h=1}^{H_n^t} \left(\nabla_{\boldsymbol{w}_s} F_n(\boldsymbol{x}_n^{t,h}) - \nabla_{\boldsymbol{w}_s} F(\boldsymbol{x}^t)\right) \right\|^2\right], \end{aligned} \tag{11}$$

where

$$\begin{aligned} & \mathbb{E}\left[\left\| \sum_{n=1}^{N} a_n \sum_{h=1}^{H_n^t} \left(\nabla_{\boldsymbol{w}_s} F_n(\boldsymbol{x}_n^{t,h}) - \nabla_{\boldsymbol{w}_s} F(\boldsymbol{x}^t)\right) \right\|^2\right] \\ \leq & N \sum_{n=1}^{N} (a_n)^2 H_n^t \sum_{h=1}^{H_n^t} \mathbb{E}\left[\|\nabla_{\boldsymbol{w}_s} F_n(\boldsymbol{x}_n^{t,h}) - \nabla_{\boldsymbol{w}_s} F(\boldsymbol{x}^t)\|^2\right] \\ \leq & N \sum_{n=1}^{N} (a_n)^2 H_n^t \sum_{h=1}^{H_n^t} \mathbb{E}\left[\|\nabla_{\boldsymbol{w}_s} F_n(\boldsymbol{x}_n^{t,h}) - \nabla_{\boldsymbol{w}_s} F_n(\boldsymbol{x}^t) + \nabla_{\boldsymbol{w}_s} F_n(\boldsymbol{x}^t) - \nabla_{\boldsymbol{w}_s} F(\boldsymbol{x}^t)\|^2\right] \\ \leq & 2N \sum_{n=1}^{N} (a_n)^2 H_n^t \sum_{h=1}^{H_n^t} (S^2 \mathbb{E}[\|\boldsymbol{x}_n^{t,h} - \boldsymbol{x}^t\|^2] + \epsilon^2) \end{aligned} \tag{12}$$

Lemma 2 (Multiple Local Training Steps in Each Round)**.** *Under Assumptions 1–3 and $\eta^t \leq \frac{1}{2S(1+H_n^t)}$, the following inequality holds:*

$$\begin{aligned} \sum_{h=1}^{H_n^t} \mathbb{E}\left[\left\|\boldsymbol{x}_n^{t,h} - \boldsymbol{x}^t\right\|^2\right] \leq & 16(H_n^t)^3 (\eta^t)^2 \left(\sigma_s^2 + \sigma_c^2 + \epsilon^2 + \mathbb{E}\left[\left\|\nabla F(\boldsymbol{x}^t)\right\|^2\right]\right) \\ & + 16(H - H_n^t)^2 (\eta^{t-\tau_n(t)+1})^2 G_c^2 + 16(\tau_n(t) - 1)^2 (\eta^{t-\tau_n(t)+1})^2 M H^2 G_c^2. \end{aligned} \tag{13}$$

Proof. We consider the following bound:

$$
\begin{aligned}
&\mathbb{E}\left[\left\|\boldsymbol{x}_n^{t,h}-\boldsymbol{x}^t\right\|^2\right]\\
=&\mathbb{E}\left[\left\|\boldsymbol{x}_n^{t,h-1}-\eta^t g_n^{t,h-1}-\boldsymbol{x}^t\right\|^2\right]\\
=&\mathbb{E}\left[\left\|\boldsymbol{x}_n^{t,h-1}-\boldsymbol{x}^t-\eta^t(g_n^{t,h-1}-\nabla F_n(\boldsymbol{x}_n^{t,h-1})+\nabla F_n(\boldsymbol{x}_n^{t,h-1})-\nabla F_n(\boldsymbol{x}^t)+\nabla F_n(\boldsymbol{x}^t)-\nabla F(\boldsymbol{x}^t)+\nabla F(\boldsymbol{x}^t)\right\|^2\right]\\
\overset{(a)}{\leq}&(1+\frac{1}{H_n^t})\mathbb{E}\left[\left\|\boldsymbol{x}_n^{t,h-1}-\boldsymbol{x}^t\right\|^2\right]\\
&+4(1+H_n^t)(\eta^t)^2\Bigg\{\mathbb{E}\left[\left\|g_n^{t,h-1}-\nabla F_n(\boldsymbol{x}_n^{t,h-1})\right\|^2\right]+\mathbb{E}\left[\left\|\nabla F_n(\boldsymbol{x}_n^{t,h-1})-\nabla F_n(\boldsymbol{x}^t)\right\|^2\right]\\
&+\mathbb{E}\left[\left\|\nabla F_n(\boldsymbol{x}^t)-\nabla F(\boldsymbol{x}^t)\right\|^2\right]+\mathbb{E}\left[\left\|\nabla F(\boldsymbol{x}^t)\right\|^2\right]\Bigg\}\\
\leq&(1+\frac{1}{H_n^t}+4(1+H_n^t)(\eta^t)^2S^2)\mathbb{E}\left[\left\|\boldsymbol{x}_n^{t,h-1}-\boldsymbol{x}^t\right\|^2\right]+4(1+H_n^t)(\eta^t)^2\left(\sigma_s^2+\sigma_c^2+\epsilon^2+\mathbb{E}\left[\left\|\nabla F(\boldsymbol{x}^t)\right\|^2\right]\right),
\end{aligned}
$$

where (a) utilizes $(X+Y)^2\leq(1+a)X^2+(1+\frac{1}{a})Y^2$ for some posive $a>0$.

By selecting $\eta^t\leq\frac{1}{2S(1+H_n^t)}$, we have:

$$
\mathbb{E}\left[\left\|\boldsymbol{x}_n^{t,h}-\boldsymbol{x}^t\right\|^2\right]\leq(1+\frac{2}{H_n^t})\mathbb{E}\left[\left\|\boldsymbol{x}_n^{t,h-1}-\boldsymbol{x}^t\right\|^2\right]+4(1+H_n^t)(\eta^t)^2\left(\sigma_s^2+\sigma_c^2+\epsilon^2+\mathbb{E}\left[\left\|\nabla F(\boldsymbol{x}^t)\right\|^2\right]\right).
$$

For simplicity, we denote:

$$
\begin{aligned}
P^{t,h}&\triangleq\mathbb{E}\left[\left\|\boldsymbol{x}_n^{t,h}-\boldsymbol{x}^t\right\|^2\right],\\
Q^t&\triangleq 4(1+H_n^t)(\eta^t)^2\left(\sigma_s^2+\sigma_c^2+\epsilon^2+\mathbb{E}\left[\left\|\nabla F(\boldsymbol{x}^t)\right\|^2\right]\right),\\
R&\triangleq 1+\frac{2}{H_n^t}.
\end{aligned}
$$

By induction, we have

$$
\begin{aligned}
P^{t,h}&\leq RP^{t,h-1}+Q^t,\\
&\leq(R)^hP^{t,0}+Q^t\sum_{j=0}^{h-1}(R)^j.
\end{aligned}
$$

Therefore, we obtain:

$$
\sum_{h=1}^{H_n^t}P^{t,h}\leq 8P^{t,0}+2(H_n^t)^2Q^t. \tag{14}
$$

We notice that:

$$
\begin{aligned}
P^{t,0}&=\mathbb{E}\left[\left\|\boldsymbol{x}_n^{t,0}-\boldsymbol{x}^t\right\|^2\right]\\
&=\mathbb{E}\left[\left\|\boldsymbol{w}_{s,n}^{t,0}-\boldsymbol{w}_s^t\right\|^2\right]+\mathbb{E}\left[\left\|\boldsymbol{w}_{c,n}^{t,0}-\boldsymbol{w}_c^t\right\|^2\right]\\
&\leq 0+\mathbb{E}\left[\left\|\boldsymbol{w}_{c,n}^{t-1,H_n^{t-1}}-\boldsymbol{w}_c^t\right\|^2\right]\\
\leq&\mathbb{E}\left[\left\|\boldsymbol{w}_{c,n}^{t-1,H_n^{t-1}}-\boldsymbol{w}_c^{t-\tau_n(t)+1}+\boldsymbol{w}_c^{t-\tau_n(t)+1}-\boldsymbol{w}_c^t\right\|^2\right]\\
\leq&2\mathbb{E}\left[\left\|\boldsymbol{w}_{c,n}^{t-1,H_n^{t-1}}-\boldsymbol{w}_c^{t-\tau_n(t)+1}\right\|^2\right]+2\mathbb{E}\left[\left\|\boldsymbol{w}_c^{t-\tau_n(t)+1}-\boldsymbol{w}_c^t\right\|^2\right].
\end{aligned} \tag{15}
$$

We assume the η^t sequence is monotonically non-increasing as t evolves. Then, we have:

$$
\begin{aligned}
\mathbb{E}\left[\left\|\boldsymbol{w}_{c,n}^{t-1,H_n^{t-1}}-\boldsymbol{w}_c^{t-\tau_n(t)+1}\right\|^2\right] \leq & (H-H_n^t)\sum_{k=t-\tau_n(t)+1}^{t-1}\sum_{h=1}^{H_n^k}(\eta^k)^2\mathbb{E}[\|g_{c,n}^{k,h}\|^2] \\
\leq & (H-H_n^t)\sum_{k=t-\tau_n(t)+1}^{t-1}\sum_{h=1}^{H_n^k}(\eta^k)^2 G_c^2 \\
\leq & (H-H_n^t)^2(\eta^{t-\tau_n(t)+1})^2 G_c^2.
\end{aligned} \tag{16}
$$

We complete the proof by combining (14)–(16). □

The last term left in the server-side error bound is:

$$
\begin{aligned}
\mathbb{E}\left[\left\|\boldsymbol{w}_c^{t-\tau_n(t)+1}-\boldsymbol{w}_c^t\right\|^2\right] \leq & (\tau_n(t)-1)\sum_{k=t-\tau_n(t)+1}^{t-1}\mathbb{E}\left[\left\|\boldsymbol{w}_c^k-\boldsymbol{w}_c^{k+1}\right\|^2\right] \\
\leq & (\tau_n(t)-1)\sum_{k=t-\tau_n(t)+1}^{t-1}(\eta^k)^2|\mathcal{M}_k|\sum_{n\in\mathcal{M}_k}(b_n^k)^2H^2G_c^2 \\
\leq & (\tau_n(t)-1)^2(\eta^{t-\tau_n(t)+1})^2MH^2G_c^2.
\end{aligned} \tag{17}
$$

The upper bound of the client-side model update is more complicated. We can derive the following upper bound with a similar technique utilized in (14):

$$
\begin{aligned}
& \mathbb{E}\left[\langle\nabla_{\boldsymbol{w}_c}F(\boldsymbol{x}^t),\boldsymbol{w}_c^{t+1}-\boldsymbol{w}_c^t\rangle\right] \\
\leq & \left\langle\nabla_{\boldsymbol{w}_c}F(\boldsymbol{x}^t),\mathbb{E}\left[-\eta^t\sum_{n\in\mathcal{M}_t}b_n^t\sum_{k=t-\tau_n(t)+1}^{t}\sum_{h=1}^{H_n^k}g_{c,n}^{k,h}\right]\right\rangle, \\
= & -\frac{\eta^tH}{2}\left\|\nabla_{\boldsymbol{w}_c}F(\boldsymbol{x}^t)\right\|^2-\frac{\eta^t}{2H}\mathbb{E}\left[\left\|\sum_{n\in\mathcal{M}_t}b_n^t\sum_{k=t-\tau_n(t)+1}^{t}\sum_{h=1}^{H_n^k}\nabla_{\boldsymbol{w}_c}F_n(\boldsymbol{x}_n^{k,h})\right\|^2\right] \\
& +\frac{\eta^t}{2H}\mathbb{E}\Big[\Big\|\sum_{n\in\mathcal{M}_t}b_n^t\sum_{k=t-\tau_n(t)+1}^{t}\sum_{h=1}^{H_n^k}\left(\nabla_{\boldsymbol{w}_c}F_n(\boldsymbol{x}_n^{k,h})-\nabla_{\boldsymbol{w}_c}F(\boldsymbol{x}^t)\right)\Big\|^2\Big],
\end{aligned} \tag{18}
$$

where

$$
\begin{aligned}
& \mathbb{E}\left[\left\|\sum_{n\in\mathcal{M}_t}b_n^t\sum_{k=t-\tau_n(t)+1}^{t}\sum_{h=1}^{H_n^k}\left(\nabla_{\boldsymbol{w}_c}F_n(\boldsymbol{x}_n^{k,h})-\nabla_{\boldsymbol{w}_c}F(\boldsymbol{x}^t)\right)\right\|^2\right] \\
\leq & |\mathcal{M}_t|\sum_{n\in\mathcal{M}_t}(b_n^t)^2H\sum_{k=t-\tau_n(t)+1}^{t}\sum_{h=1}^{H_n^k}\mathbb{E}\left[\left\|\left(\nabla_{\boldsymbol{w}_c}F_n(\boldsymbol{x}_n^{k,h})-\nabla_{\boldsymbol{w}_c}F(\boldsymbol{x}^t)\right)\right\|^2\right] \\
\leq & |\mathcal{M}_t|\sum_{n\in\mathcal{M}_t}(b_n^t)^2H\sum_{k=t-\tau_n(t)+1}^{t}\sum_{h=1}^{H_n^k} \\
& \mathbb{E}\left[\|\nabla_{\boldsymbol{w}_c}F_n(\boldsymbol{x}_n^{k,h})-\nabla_{\boldsymbol{w}_c}F_n(\boldsymbol{x}^k)+\nabla_{\boldsymbol{w}_c}F_n(\boldsymbol{x}^k)-\nabla_{\boldsymbol{w}_c}F_n(\boldsymbol{x}^t)+\nabla_{\boldsymbol{w}_c}F_n(\boldsymbol{x}^t)-\nabla_{\boldsymbol{w}_c}F(\boldsymbol{x}^t)\|^2\right] \\
\leq & 3|\mathcal{M}_t|\sum_{n\in\mathcal{M}_t}(b_n^t)^2H\sum_{k=t-\tau_n(t)+1}^{t}\sum_{h=1}^{H_n^k}(S^2\mathbb{E}[\|\boldsymbol{x}_n^{k,h}-\boldsymbol{x}^k\|^2]+S^2\mathbb{E}[\|\boldsymbol{w}_s^k-\boldsymbol{w}_s^t\|^2]+S^2\mathbb{E}[\|\boldsymbol{w}_c^k-\boldsymbol{w}_c^t\|^2]+\epsilon^2).
\end{aligned} \tag{19}
$$

We first derive the upper bound for $\mathbb{E}[\|\boldsymbol{x}_n^{k,h}-\boldsymbol{x}^k\|^2]$. According to Lemma 2, we obtain:

$$
\begin{aligned}
\sum_{k=t-\tau_n(t)+1}^{t}\sum_{h=1}^{H_n^k}\mathbb{E}[\|\boldsymbol{x}_n^{k,h}-\boldsymbol{x}^k\|^2] \leq & \sum_{k=t-\tau_n(t)+1}^{t}\Big\{16(H_n^k)^3(\eta^k)^2\left(\sigma_s^2+\sigma_c^2+\epsilon^2+\mathbb{E}\left[\left\|\nabla F(\boldsymbol{x}^k)\right\|^2\right]\right) \\
& +16(H-H_n^k)^2(\eta^{k-\tau_n(k)+1})^2G_c^2+16(\tau_n(k)-1)^2(\eta^{k-\tau_n(k)+1})^2MH^2G_c^2\Big\}.
\end{aligned} \tag{20}
$$

In terms of $\mathbb{E}\left[\|\boldsymbol{w}_s^k - \boldsymbol{w}_s^t\|^2\right]$, we have:

$$\begin{aligned}
\sum_{k=t-\tau_n(t)+1}^{t} \sum_{h=1}^{H_n^k} \mathbb{E}\left[\|\boldsymbol{w}_s^k - \boldsymbol{w}_s^t\|^2\right] &= \sum_{k=t-\tau_n(t)+1}^{t} H_n^k \, \mathbb{E}\left[\|\boldsymbol{w}_s^k - \boldsymbol{w}_s^t\|^2\right] \\
&\leq \sum_{k=t-\tau_n(t)+1}^{t} H_n^k \left((\tau_n(t)-1) \sum_{j=k}^{t-1} \mathbb{E}\left[\|\boldsymbol{w}_s^{j+1} - \boldsymbol{w}_s^j\|^2\right] \right) \\
&\leq (\tau_n(t)-1) H \sum_{j=t-\tau_n(t)+1}^{t-1} \mathbb{E}\left[\|\boldsymbol{w}_s^{j+1} - \boldsymbol{w}_s^j\|^2\right] \\
&\leq (\tau_n(t)-1)^2 H^3 (\eta^{t-\tau_n(t)+1})^2 N G_s^2.
\end{aligned} \tag{21}$$

Similarly, we derive:

$$\sum_{k=t-\tau_n(t)+1}^{t} \sum_{h=1}^{H_n^k} \mathbb{E}\left[\|\boldsymbol{w}_c^k - \boldsymbol{w}_c^t\|^2\right] \leq (\tau_n(t)-1)^2 H^3 (\eta^{t-\tau_n(t)+1})^2 M G_c^2. \tag{22}$$

Combining Lemma 1, 2, and inequalities (11)–(12), (17)–(22), we obtain

$$\begin{aligned}
\mathbb{E}\big[F(\boldsymbol{x}^{t+1})\big] \leq \mathbb{E}\big[F(\boldsymbol{x}^t)\big] &- \frac{\eta A_t}{2} \mathbb{E}\big[\|\nabla F(\boldsymbol{x}^t)\|^2\big] \\
&+ \eta^3 C_1 \sum_{q=t-\tau_{\max}+1}^{t} \mathbb{E}\big[\|\nabla F(\boldsymbol{x}^q)\|^2\big] \\
&+ \eta^2 \Big(C_1(\sigma_s^2 + \sigma_c^2 + \epsilon^2) + C_2 G_c^2 + C_3 G_s^2 \Big) + \eta \Big(\frac{NH^2}{A_{\min}} + \frac{3MH}{2} \Big) \epsilon^2,
\end{aligned} \tag{23}$$

where

$$A_t \triangleq \sum_{n=1}^{N} a_n H_n^t, \qquad A_{\min} \triangleq \min_{0 \leq t \leq T-1} A_t,$$

and

$$\begin{aligned}
C_1 &\triangleq \frac{16NS^2H^4}{A_{\min}} + 24MS^2\tau_{\max}H^3, \\
C_2 &\triangleq \frac{16NH^3(1+\tau_{\max}^2 M)}{A_{\min}} + 24MS^2\tau_{\max}H^2 + 24M^2S^2\tau_{\max}^3H^2 \\
&\quad + 24M^2S^2\tau_{\max}^2H^3 + \frac{S}{2}MH^2 + \frac{NH^2}{A_{\min}} + \frac{3}{2}MH, \\
C_3 &\triangleq 24MNS^2\tau_{\max}^2H^3 + \frac{S}{2}NH^2.
\end{aligned}$$

Summing (23) from $t = 0$ to $T-1$ and using the bounded-overlap inequality $\sum_{t=0}^{T-1} \sum_{q=t-\tau_{\max}+1}^{t} \phi_q \leq \tau_{\max} \sum_{t=0}^{T-1} \phi_t$, we obtain:

$$\begin{aligned}
\mathbb{E}\big[F(\boldsymbol{x}^T)\big] \leq \mathbb{E}\big[F(\boldsymbol{x}^0)\big] &- \eta \left(\frac{A_{\min}}{2} - \eta^2 \tau_{\max} C_1 \right) \sum_{t=0}^{T-1} \mathbb{E}\big[\|\nabla F(\boldsymbol{x}^t)\|^2\big] \\
&+ T\eta^2 \Big(C_1(\sigma_s^2 + \sigma_c^2 + \epsilon^2) + C_2 G_c^2 + C_3 G_s^2 \Big) + T\eta \Big(\frac{NH^2}{A_{\min}} + \frac{3MH}{2} \Big) \epsilon^2.
\end{aligned} \tag{24}$$

Therefore, we can establish the following non-convex convergence result. We select the learning rate η by:

$$\eta \leq \min \left\{ \frac{1}{2S(1+H)}, \sqrt{\frac{A_{\min}}{4\tau_{\max} C_1}} \right\},$$

then

$$\frac{A_{\min}}{2} - \eta^2 \tau_{\max} C_1 \geq \frac{A_{\min}}{4}.$$

Using (24) and the lower boundedness $F(\boldsymbol{x}^T) \geq F^\star$, we obtain

$$\frac{\eta A_{\min}}{4} \sum_{t=0}^{T-1} \mathbb{E}\big[\|\nabla F(\boldsymbol{x}^t)\|^2\big] \leq F(\boldsymbol{x}^0) - F^\star + T\eta^2 \Big(C_1(\sigma_s^2 + \sigma_c^2 + \epsilon^2) + C_2 G_c^2 + C_3 G_s^2 \Big) + T\eta(\frac{NH^2}{A_{\min}} + \frac{3MH}{2})\epsilon^2. \tag{25}$$

Dividing both sides by $\frac{\eta T A_{\min}}{4}$ yields the theorem.

Under the conditions of this theorem, if the constant stepsize is chosen as $\eta = \Theta(1/\sqrt{T})$. Then, we obtain:

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}\big[\|\nabla F(\boldsymbol{x}^t)\|^2\big] = \mathcal{O}(1/\sqrt{T}) + \mathcal{O}(\epsilon^2).$$

This completes the proof.

Remark: The estimate in (21) is conservative for simplicity. A tighter bound can be obtained by explicitly invoking the lemma in the main text, which captures the reduced cumulative server-side drift under partial aggregation. Nevertheless, even with the conservative treatment adopted here, the final convergence bound already reflects the theoretical benefit of partial server-side aggregation.

More specifically, if the server-side model were also aggregated in a ready-triggered manner, then the server-side term would follow the same stale-start accumulation argument as the client-side term. Under the same derivation logic, the coefficient of the G_s^2 term would revert from C_3 to C_2. Moreover, the coefficient C_1 multiplying $\sigma_s^2 + \sigma_c^2 + \epsilon^2$ would also become larger, since the corresponding server-side discrepancy term would accumulate over a stale interval rather than being controlled round by round. Therefore, partial server-side aggregation improves not only the server-side second-moment constant but also the constant associated with the stochastic gradient and heterogeneity terms.

References

[1] P. Han, C. Huang, G. Tian, M. Tang, and X. Liu, "Convergence analysis of split federated learning on heterogeneous data," in *Advances in Neural Information Processing Systems*, A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, Eds., vol. 37. Curran Associates, Inc., 2024, pp. 103 476–103 544. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2024/file/bb63841e1ad12370a34504f15c60db4f-Paper-Conference.pdf